

Question Number	Answer	Additional guidance	Mark
8(a)	An explanation that includes three of the following points: - determining {DNA base sequences / genome sequence} for whitefly (1) (data collated) using bioinformatics (1) - compare {DNA base sequence / genome} of whitefly with genomes of plants to see if they are the same (1) {BtPMT1 sequence / gene} was only found in plants and not in other species of whitefly (1) Alternative - use of PCR (1) - cut DNA using same restriction enzyme (1) - use of gel electrophoresis (1) - show bands from gel electrophoresis of whitefly with genomes of plants are the same (1)	please don't mix and match ACCEPT use of (DNA) microarrays ACCEPT compare database of whitefly and plant DNA to see if they are the same	(3)

Question Number	Answer	Additional guidance	Mark
8(b)	An explanation that includes four of the following points: • {BtPMT1 / the gene} allows whitefly to avoid being poisoned (when it eats the plant) (1) • accept having this gene gives them a (selective) advantage(1) • (only) whitefly with the {BtPMT1 / gene} will survive (1) • to (reproduce) and pass {BtPMT1 /gene / advantageous allele} to offspring (1) • (as a result) these whitefly have no competition for this food source (from other insects) (1)	ACCEPT allele for gene anywhere ACCEPT flies without the gene will {die / be poisoned} (if they eat the plant) ACCEPT the whiteflies can eat the plants without being poisoned ACCEPT correct ref to selection pressure ACCEPT reach the age of reproduction	(4)

Question Number	Answer	Additional guidance	Mark
8(c)	An explanation that includes four of the following points: • {isolate / cut} gene from coral (1) • insert gene into {zebrafish / embryos / zygotes} (1) • suitable method to insert gene (inserted into zebrafish) (1) • shine UV light on (adult) fish (as only the modified fish will glow) (1) • {red /glowing red / shining / fluorescing} fish are selected for breeding (1)	ignore cDNA e.g. vector, plasmid, micropipette, heat shock, liposomes, gene gun ACCEPT remove the fish that don't glow red	(4)

Question Number	Answer	Additional guidance	Mark
8(d)	An answer that includes three of the following points:	Maximum two marks for benefits Maximum two marks for risks ACCEPT allele for gene	
B	Benefits (max 2) • medical research (1) • study tissue regeneration (1) • track how skin cells move (1) • unique pets (1) • financial benefit of selling (1)	ACCEPT study cell regeneration / differentiation	
R	Risks (max 2) • the fish are more visible to predators / negative effect on ecosystem (1) • the gene affecting the health of the zebrafish (1) • the gene getting into other {populations of zebrafish / animals / species} (1) • the gene getting into (marine) food chains (1) • affecting the health of the coral (1) • encouraging illegal pet trade (1)	ACCEPT tourism at aquarium ACCEPT do not know long term effects e.g. mating with other species / crossbreeding / horizontal gene transfer	(3)

Question Number	Answer	Additional guidance	Mark
8(e)	An answer that includes three of the following points: • increase in {volume / size of cytoplasm / membrane size} (1) • (due to) increase in permeability of membrane to water (1) • (more) water entered by osmosis (to increase cell size) (1) • increased blood flow to the wound (1) • inflammatory response (1) • resulting in {cell division / mitosis} (1)	ACCEPT inflammation occurs	(3)

